1 Introduction

1.1 Arithmetic in [0,infty]

1.2 Metric Space

Lemma 1.2.1

X is a metric space, $x \in X$, B_1, B_2 are twos balls in X. If $x \in B_1 \cap B_2$, then x is the center of an open ball $B \subseteq B_1 \cap B_2$.

Proof. We may assume that B_1, B_2 have different center. Let $y_1 \neq y_2$,

$$B_1 = \{x : d(x, y_1) < r_1\}, \quad B_2 = \{x : d(x, y_2) < r_2\}$$

Take $x \in B_1 \cap B_2$, we claim that there exists $r_0 > 0$ such that $B := \{y : d(x, y) < r_0\} \subseteq B_1 \cap B_2$. Otherwise , for all $r > 0$, there exists a point y_r such that the following two hold at the same time

1. $d(x, y_r) < r$
2. $d(y_r, y_1) \geq r_1$ or $d(y_r, y_2) \geq r_2$.

Thus

$$d(x, y_1) \geq d(y_r, y_1) - d(x, y_r) > r_1 - r, \quad \text{or} \quad d(x, y_2) > r_2 - r$$

The above shows that

$$r > \min\{r_1 - d(x, y_1), r_2 - d(x, y_2)\}, \quad \forall r > 0$$

which is a contradiction. □

1.3 Limits of Set Sequences

Definition 1.3.1 ▶ Limit Inferior and Limit Superior

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. By the ***Limit Inferior of the sequence***, we mean

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \{x : \exists N_0 \in \mathbb{N}, \forall n > N_0, x \in A_n\}$$

2. By the ***Limit Superior of the sequence***,we mean

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \{x : \forall N \in \mathbb{N}, \exists n \geq N, x \in A_n\}$$

Note.

1. Limit Inferior 包含那些“最终稳定下来”的元素，即从某个点之后就永远属于序列中所有后续集合的元素。
2. Limit Superior 包含那些“反复出现”的元素，即在无限多个集合中出现的元素。

Proposition 1.3.2

For any sequence of sets $\{A_n\}$, it always holds that:

$$\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$$

Proof sketch.
直觉上是显然的，因为“最终稳定下来”的元素一定会“反复出现”.

Proof. It is obvious by the $\{x : x \in P\}$ form representation of the Limit Inferior and Superior. □

Definition 1.3.3 ▶ Limit of a Set Sequence

If the limit inferior and limit superior of a set sequence $\{A_n\}_{n=1}^{\infty}$ are equal, i.e., $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$, then we say the ***limit*** of the sequence exists, and it is defined as:

$$\lim_{n\to\infty} A_n = \liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$$

Proposition 1.3.4

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. If $\{A_n\}$ is an increasing sequence, then the limit of $\{A_n\}$ exists and
$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$
2. If $\{A_n\}$ is a decreasing sequence, then the limit of $\{A_n\}$ exists and
$$\lim_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

Proof. 1. If $\{A_n\}$ is increasing , then

$$\bigcap_{n=N}^{\infty} A_n = A_N$$

and

$$\bigcup_{n=N}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n, \forall N \in \mathbb{N}.$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty} A_N$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

which are the same.

2. If $\{A_n\}$ is decreasing, then

$$\bigcap_{n=N}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\bigcup_{n=N}^{\infty} A_n = A_N$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} A_N$$

which are the same.

□

Proposition 1.3.5

For a sequence of sets $\{A_n\}_{n=1}^{\infty}$ and their corresponding sequence of indicator functions $\{\chi_{A_n}(x)\}_{n=1}^{\infty}$:

- $\chi_{\liminf_{n\to\infty} A_n}(x) = \liminf_{n\to\infty} \chi_{A_n}(x)$
- $\chi_{\limsup_{n\to\infty} A_n}(x) = \limsup_{n\to\infty} \chi_{A_n}(x)$
- $\lim_{n\to\infty} A_n$ exists, then $\chi_{\lim_{n\to\infty} A_n}(x) = \lim_{n\to\infty} \chi_{A_n}(x)$

Proof sketch.

- 若 x 在 limit inferior 里面，则 x 是“最终稳定”的，$\chi_{A_n}(x)$ 是关于 n “最终”恒为 1 的.
- 若 x 在 limit superior 里面，则 x 是“反复出现”的，即相当于 N 多大，总会出现之后的某个 n 使得 $\chi_{A_n}(x) = 1$.
- 当极限存在时，函数列 $\left\{\chi_{A_n}(x)\right\}_{n=1}^{\infty}$ 的 limit inferior 和 supperior 根据上两条相等，等于极限集合的 χ.

1.4 Exercises

Problem 1.4.1 ▶ Convex Bodies

A convex body is a compact, convex subset $K \subseteq \mathbb{R}^n$ which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on $\mathbb{R}^n$:

$$N(x) := \inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\}$$

2. Prove that K is homeomorphic to $\mathbb{D}^n$ and that ∂K is homeomorphicc to $\mathbb{S}^{n-1}$.

Note.

1. 若 K 是一个单位球. 则 $N(x) = \|x\|$.
2. 证明 $N(x)$ 是一个 norm, 需要证明:1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于 φ 的形式足够简单, 可以通过直接构造逆映射证明双射.

Proof. 1. It is obvious that $N(x) \geq 0$. If $N(x) = 0$, then

$$\inf\left\{\lambda > 0 : \frac{x}{\lambda} \in K\right\} = 0$$

which means that for all $\varepsilon > 0$, $\frac{x}{\varepsilon} \in K$ .Since K is compact, $\operatorname{dist}(K, 0) < \infty$. If $x \neq 0$,then $\|x\| > 0$. We take $\varepsilon_0 < \frac{1}{\operatorname{dist}(K,0)\cdot\|x\|}$. Then $\left\|\frac{x}{\varepsilon_0}\right\| > \operatorname{dist}(K, 0)$. But $\frac{x}{\varepsilon_0} \in K$, which is a contradiction.For the second part, let $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$. For all $\lambda > N(x)$,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

,which shows that $N(kx) \leq kN(x)$. Replace k by $\frac{1}{k}$ and replace x by kx, we have $N(kx) \geq kN(x)$. Thus $N(kx) = kN(x)$. For the triangle inequality , we need to show that

$$\inf\left\{\lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K\right\} \leq \inf\left\{\lambda > 0 : \frac{x_1}{\lambda} \in K\right\} + \inf\left\{\lambda > 0 : \frac{x_2}{\lambda} \in K\right\}$$

Let $\lambda_0 = N(x_1 + x_2)$. Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all $\delta > 0$, we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since K is convex , we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta}\left((N(x_2) + \delta)\frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta)\frac{x_1}{N(x_1) + \delta}\right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all $\delta > 0$. Then

$$N(x_1 + x_2) \le N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \le N(x_1) + N(x_2)$$

2. $x \in K^\circ \iff N(x) < 1$, $x \in \operatorname{Ext}(K) \iff N(x) > 1$, $x \in \partial K \iff N(x) = 1$. We define a map $\varphi : \mathbb{R}^n \backslash \to \mathbb{R}^n$.

$$\varphi(x) := \begin{cases} N(x)\frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then φ maps K° on to $(\mathbb{D}^n)^\circ$, and maps ∂K on to $\mathbb{S}^{n-1}$. To show φ is continious , only need to show it is continious at 0 . In fact, $\|\lim_{x\to 0}\varphi(x)\| = \lim_{x\to 0}\|\varphi(x)\| = \lim_{x\to 0} N(x) = 0$ since N is a norm . We define $\psi : \mathbb{R}^n \to \mathbb{R}^n$:

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)}x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then $\varphi \circ \psi = \psi \circ \varphi = \mathrm{Id}$. The same skills can be used to show ψ is continious at 0 as well . Finally , since $\mathbb{S}^{n-1}$ is a subspace of $\mathbb{D}^n$, $\psi|_{\mathbb{S}^{n-1}} : \mathbb{S}^{n-1} \to \psi\left(\mathbb{S}^{n-1}\right) = \partial K$ is also a homeomorphic .

$\square$

Problem 1.4.2

Let X, X' be topological spaces and $x \in X$, $x' \in X'$ be base points. Prove that the wedge sum $X \vee X'$ is homeomorphic to the subspace:

$$(X \times \{x'\}) \cup (\{x\} \times X') \subseteq X \times X'$$